

Supernovae Quiz

Cosmic Compass

November 19, 2025

Multiple-Choice Questions

1. What is the primary cause of a Type II supernova?

- A. Collapse of a white dwarf
- B. Collapse of a massive star
- C. Explosion of a neutron star
- D. Merger of two black holes

Answer: B. Collapse of a massive star.

Explanation: Type II supernovae occur when a massive star exhausts its nuclear fuel and collapses under its own gravity, leading to a catastrophic explosion.

2. Which of the following is a remnant of a supernova explosion?

- A. White dwarf
- B. Neutron star
- C. Red giant
- D. Brown dwarf

Answer: B. Neutron star.

Explanation: Neutron stars are often formed from the remnants of supernova explosions, where the core of the star collapses into an extremely dense object.

3. How often does a supernova occur in the Milky Way galaxy?

- A. Every year
- B. Every 25 to 100 years
- C. Every 1000 years
- D. Every 10,000 years

Answer: B. Every 25 to 100 years.

Explanation: On average, a supernova explosion occurs in the Milky Way galaxy every 25 to 100 years, though many go unnoticed due to dust blocking the view.

4. What is the difference between a nova and a Type Ia supernova?
- A. A nova involves a white dwarf, while a Type Ia supernova involves a neutron star.
 - B. A nova is a smaller explosion, while a Type Ia supernova is a catastrophic explosion of a white dwarf.
 - C. A nova occurs in binary systems, while a Type Ia supernova occurs in single stars.
 - D. There is no difference; they are the same phenomenon.

Answer: B. A nova is a smaller explosion, while a Type Ia supernova is a catastrophic explosion of a white dwarf.

Explanation: A nova involves a temporary increase in brightness due to material accreting onto a white dwarf, whereas a Type Ia supernova results from the complete destruction of a white dwarf.

5. Which of the following elements is primarily produced in supernova explosions?
- A. Hydrogen
 - B. Helium
 - C. Iron
 - D. Gold

Answer: D. Gold.

Explanation: Heavy elements like gold are primarily produced in the extreme conditions of supernova explosions.

6. What is the Crab Nebula an example of?
- A. A planetary nebula
 - B. A supernova remnant
 - C. A black hole
 - D. A protostar

Answer: B. A supernova remnant.

Explanation: The Crab Nebula is the remnant of a supernova explosion observed in 1054 AD.

7. Which type of supernova is used as a "standard candle" in astronomy?
- A. Type I

- B. Type II
- C. Type Ia
- D. Type Ib

Answer: C. Type Ia.

Explanation: Type Ia supernovae have a consistent peak luminosity, making them useful for measuring cosmic distances.

8. What is the fate of a star with an initial mass of 8 solar masses?
- A. It becomes a white dwarf.
 - B. It becomes a neutron star.
 - C. It becomes a black hole.
 - D. It explodes as a Type Ia supernova.

Answer: B. It becomes a neutron star.

Explanation: Stars with initial masses between 8 and 20 solar masses typically end their lives as neutron stars after a supernova explosion.

9. What is the primary energy source for the radiation emitted by the Crab Nebula?
- A. Nuclear fusion
 - B. Gravitational contraction
 - C. The pulsar at its center
 - D. Dark matter

Answer: C. The pulsar at its center.

Explanation: The Crab Nebula's radiation is powered by the rapidly spinning neutron star (pulsar) at its center.

10. Which of the following is NOT a consequence of a nearby supernova explosion?
- A. Formation of heavy elements
 - B. Destruction of nearby planets
 - C. Creation of a black hole
 - D. Increase in cosmic background radiation

Answer: D. Increase in cosmic background radiation.

Explanation: Supernovae do not affect the cosmic background radiation, which is a remnant of the Big Bang.

Short Answer Questions

1. Describe the differences between Type Ia and Type II supernovae.

Answer: Type Ia supernovae result from the thermonuclear explosion of a white dwarf in a binary system, while Type II supernovae occur when a massive star collapses under its own gravity after exhausting its nuclear fuel.

2. Why are supernovae important for the chemical enrichment of the universe?

Answer: Supernovae produce and distribute heavy elements like iron, gold, and uranium into space, which are essential for the formation of planets and life.

3. Explain why neutron stars are often found in supernova remnants.

Answer: Neutron stars are the dense cores left behind after a massive star undergoes a supernova explosion, making them common in supernova remnants.

4. What role do supernovae play in the evolution of galaxies?

Answer: Supernovae regulate star formation by dispersing gas and dust, and they contribute to the chemical enrichment of galaxies, influencing future generations of stars.

5. How did the supernova observed in 1006 AD differ from typical supernovae?

Answer: The supernova of 1006 AD was exceptionally bright, visible during the day, and recorded by observers worldwide, making it one of the most dramatic supernovae in history.

Visual Aids

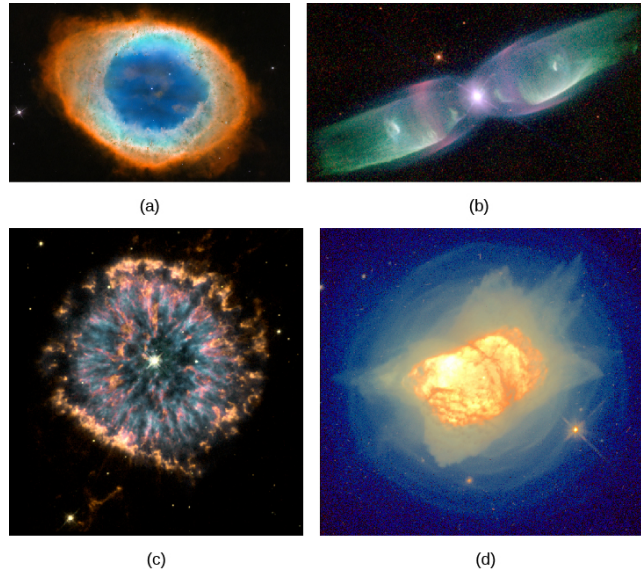


Figure 1: Supernova remnants in other galaxies.



Figure 2: Illustration of a supernova explosion.

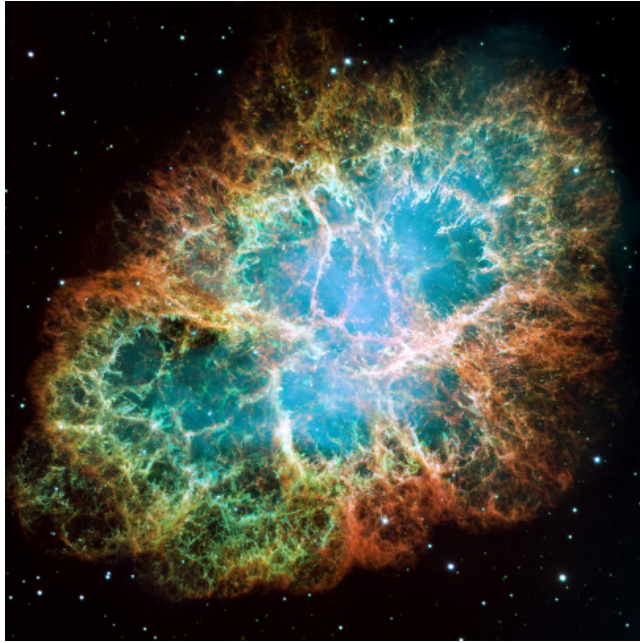


Figure 3: The Crab Nebula, a famous supernova remnant.